

Key Points:

- Dipole structures emerge as a dominant pattern of vertical velocity within mesoscale eddies in numerical model
- Opposite dipole structures are observed between the subtropical and the western boundary current extension regions
- The β -effect and deformation effect associated with background flows modulate opposite dipole structures

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Dipole Structure of Vertical Velocity Induced by Mesoscale Eddies

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Abstract Mesoscale eddies typically maintain a state of quasi-geostrophic balance and are accompanied by vertical motions throughout their lifecycles. However, the spatial structures and underlying mechanisms driving these eddy-induced vertical velocities remain unclear. Using a $1/32^\circ$ idealized basin numerical simulation, we investigate the vertical velocity structures of mesoscale eddies and their governing mechanisms, focusing on the contrasting features between the subtropical region and western boundary current extension. A statistical analysis on the numerical results shows that dipole structure is the predominant vertical velocity pattern, accounting for approximately 65% of all snapshots, which is much higher than the occurrence of monopole and quadrupole structures. Furthermore, composite analyses reveal zonally opposite patterns of dipole structures between the two regions with different background flows. Vorticity diagnostics demonstrate that the β -effect dominates the dipole structures in the subtropical region, whereas in the extension region the enhanced ageostrophic effect dominates the dipole patterns. Additional analysis using the quasi-geostrophic omega equation indicates that the opposite dipole structures result from the combination of the β -effect and horizontal deformation effect related to the influence of eddy-flow interaction. These findings provide novel insights into the dynamics of vertical velocity in mesoscale eddies and may advance our understanding of biogeochemical processes across different regions.

Plain Language Summary Most previous studies on eddy-induced vertical velocities neglect the regional dynamic regime and the β -effect associated with the meridional variation of planetary vorticity. Using a $1/32^\circ$ idealized basin numerical simulation, we found that the dipole structure is the predominant pattern of eddy-induced vertical velocity. The dipole structure is dominated by the β -effect in the subtropical region whereas an opposite dipole pattern in the extension region is influenced by the strong background current related to the meridional density variations. These results advance understanding of mesoscale eddy dynamics and their implications for regional biogeochemical variability.

1. Introduction

Mesoscale eddies, characterized by spatial scales of O (10–100 km) and temporal scales ranging from weeks to months, are ubiquitous features in the global ocean (Chaigneau et al., 2011). They carry nearly 90% of the ocean's kinetic energy and play a crucial role in material transport in the upper ocean (Chapman et al., 2024; Ferrari & Wunsch, 2009; Gaube et al., 2013; Griffies et al., 2015; Liu et al., 2024; Qu et al., 2022). Specifically, upwelling within eddies promotes primary production by uplifting nutrients into the euphotic layer, especially in the oligotrophic open ocean (Barceló-Llull et al., 2016; Falkowski et al., 1991; Klein & Lapeyre, 2009; Pascual et al., 2015; Pidcock et al., 2016; Wang et al., 2021; Zhang et al., 2019). Conversely, downwelling accelerates the sinking of phytoplankton detritus, contributing to sedimentation processes and the global carbon cycle (Gruber et al., 2011; Lathuiliere et al., 2010; van Haren et al., 2006; Zhang et al., 2015). Thus, vertical motions induced by mesoscale eddies play a significant role in modulating biogeochemical processes and the global climate system (McGillicuddy et al., 1998; McGillicuddy & Robinson, 1997).

Early studies on eddy-induced vertical motions suggested a monopole pattern associated with the vertical movement of isopycnal in the eddy center, explained by the “eddy pumping” and linear “eddy-Ekman pumping” mechanisms (Gaube et al., 2015; McGillicuddy et al., 2007; McGillicuddy & Robinson, 1997). Beyond the conventional monopole structure, growing evidence points to the existence of multipole vertical velocity

structures within mesoscale eddies. For example, Mason et al. (2017) revealed dipole structures aligned with the eddy propagation orientation through composite analyses in the western boundary current system. Similar dipole patterns, rotating around the eddy center, were identified within an anticyclonic eddy near the Canary Islands (Estrada-Allis et al., 2019). Recently, Hou et al. (2022) and Qu et al. (2022) reported dipole structures aligned with the zonal and meridional directions, respectively. These findings suggest that dipole patterns may manifest in various forms across different regions. Meanwhile, several studies have also identified quadrupole structures (e.g., Bashmachnikov et al., 2018; Chen et al., 2020; Pilo et al., 2018) and even more complex patterns (e.g., Claret & Viúdez, 2010), offering different perspectives on eddy-induced vertical velocity. However, there is a lack of consensus regarding the dominant pattern and relative percentages of the different structures.

Alongside the identification of these various complex structures, a range of physical mechanisms have also been proposed for the generation of vertical motions within mesoscale eddies. For example, the nonlinear Ekman pumping and eddy-induced sea surface temperature (SST)-wind interaction have been proposed as mechanisms complementary to classical Ekman pumping (Chelton & Xie, 2010; Gaube et al., 2015; McGillicuddy et al., 2008). Beyond the simplified conceptual models, various mechanisms have been suggested with continuous study including vortex Rossby waves (e.g., McWilliams et al., 2003; Nardelli, 2013), eddy distortion (e.g., Pilo et al., 2018), vortex tilting (e.g., Li et al., 2022; Viúdez, 2018), the influence of submesoscale and ageostrophic dynamics (e.g., Barceló-Llull et al., 2017; Brannigan, 2016; Zhang et al., 2019), and the decaying phase of the eddy lifecycle (e.g., Chen et al., 2020).

Among these mechanisms, the role of ageostrophic dynamics has received increasing attention in recent studies. Several works using quasi- or semi-geostrophic omega equations have examined the influence of ageostrophic advection of vorticity and straining deformation. For example, Barceló-Llull et al. (2017) proposed that the dipole structure results from a combination of horizontal deformation and the advection of vertical relative vorticity by ageostrophic vertical shear. Moreover, there are also studies adopt the generalized Omega equation in examining the effect of ageostrophic dynamics on vertical velocity. For instance, Uchida et al. (2019) decompose the vertical transport into balanced and unbalanced components. They suggest that the mesoscale vertical fluxes provide a first-order pathway for the vertical iron transport at depths below the mixed layer, whereas within the mixed layer, the relative contribution by unbalanced motion increases significantly. Using the generalized omega equation, Qiu et al. (2020) found that the majority of the vertical velocity is determined by kinematic deformation of density anomalies by geostrophic horizontal motions below the surface mixed layer. Pietri et al. (2021) point out that ageostrophic forcing terms due to flow deformation are substantial where near-surface frontal processes are important. Estrada-Allis et al. (2019) further suggested that the dipole structure arises from an imbalance between the advection and stretching of the vorticity forcing terms. Similarly, Chen et al. (2020) demonstrated a comparable magnitude of Ekman pumping and straining deformation in nutrient flux. However, most of these studies are conducted within the f -plane framework, thereby neglecting the influence of the β -effect. Hou et al. (2022) highlighted the importance of the β -effect in shaping the dipole structure by conducting a series of numerical experiments with a single idealized eddy in a stationary basin, with and without the β -effect. Yet, they omitted the role of deformation effect associated with the background flow, leaving the relative importance of these two mechanisms unresolved. These findings suggest that, despite numerous proposed explanations, a consensus regarding the mechanisms governing multipole vertical velocity structures has yet to be established.

The objective of this study is to investigate the general features of the above-mentioned dipole structures induced by mesoscale eddies and the dynamic mechanisms, with a particular focus on comparisons between the subtropical region and the western boundary current extension. We conduct a series of statistics and composite analysis on an idealized high-resolution numerical simulation. The paper is organized as follows. In Section 2, we present a description of the idealized model and the eddy detection algorithm. In Section 3, the general features of vertical velocity structures are compared, and a composite analysis is conducted, followed by a diagnostic analysis to explore the governing mechanisms. A discussion of the results and the concluding remarks are presented in Section 4.

2. Data and Methods

2.1. High-Resolution Numerical Simulation Output

To generate materially coherent eddies (i.e., eddies with long-lived concentrations of energy and vorticity Koszalka et al., 2009) with characteristics broadly similar to those of real ocean and investigate the dynamic

mechanisms of vertical velocity within them, we employ an idealized high-resolution ($1/32^\circ$) numerical model configured by Liu et al. (2022) based on the Massachusetts Institute of Technology general circulation model (MITgcm; Marshall et al., 1997). The model domain consists of an idealized rectangular basin spanning 60° in both longitude and latitude in spherical polar coordinates, with a flat bottom at a depth of 4,000 m. The southern boundary of the domain coincides with the equator, and the western/eastern boundary aligned with longitudes to avoid unnecessary complications of the topographic effect. The model horizontal resolution is set to $1/32^\circ$ (~ 3 km), and the vertical discretization consists of 29 z-levels. The upper 1,000 m includes 20 levels, with vertical spacings gradually increasing downward from 10 to 100 m. No-slip conditions are applied at all four solid boundaries. Additionally, a continental shelf is imposed along the western boundary, obeying an exponential profile from 200 m at the west to 4,000 m in 4° to the east.

The model adopts a linear equation of state, with density depending only on the temperature:

$$\rho = \rho_{\text{ref}} + 1000[-\alpha_T(T - 20)], \quad (1)$$

where the reference density ρ_{ref} is $1,025 \text{ kg m}^{-3}$, and the thermal expansion coefficient α_T is set to $2 \times 10^{-4} \text{ }^\circ\text{C}^{-1}$.

The model is forced by a zonally uniform wind stress (see W0 of Figure 1a in Liu et al. [2022]), with the latitudes of maximum wind stress and wind stress curl around 36°N and 24°N , respectively, to represent the actual wind field of the subtropical gyre. To simulate the differential surface heating and the strong subtropical front without giving radiative fluxes, the surface potential density is restored and a density “jump” is set in our model, aligning with the maximum eastward wind stress (see Figure 1b in Liu et al. [2022]). The biharmonic viscosity is calculated according to the velocity gradient based on the modified Leith scheme (Fox-Kemper & Menemenlis, 2008), while the lateral heat diffusivity is set to zero. The vertical viscosity A_v is set to $1 \times 10^{-4} \text{ m}^2 \text{ s}^{-1}$, and the diapycnal diffusivity K_v is $1 \times 10^{-7} \text{ m}^2 \text{ s}^{-1}$. These configurations successfully drive a stable and realistic subtropical circulation, which in turn generates an energetic mesoscale eddy field through large-scale instability processes. The validity of this model configuration has been well-established in earlier studies (e.g. Liu et al., 2019). Furthermore, this energetic eddy field effectively resolves mesoscale dynamics and the associated vertical motions, and have been used for discussing the eddy-flow interactions (Liu et al., 2019, 2022). Further comprehensive details regarding the model configuration and its validation are provided in Liu et al. (2022).

The model is integrated for 50 years to allow the upper-ocean circulation to reach equilibrium and then continued to run another 1 year for subsequent analysis, during which the daily-averaged model outputs of sea surface height (SSH), three-dimensional velocity, and temperature are saved for eddy identification, tracking, and the subsequent analysis.

Our analyses focus on two regions with higher eddy activity exhibiting distinct dynamic features: the western boundary current extension (25°N – 45°N , 5°E – 55°E , hereafter, the extension region) and the subtropical countercurrent (5°N – 25°N , 5°E – 55°E , hereafter, the subtropical region) (Figure 1a), to compare the general characteristics of eddy vertical velocities between them. The general patterns of temperature gradient and zonal velocity in the two regions show differences (Figures 1b and 1c). Specifically, the extension region features a strong eastward current within the upper 500 m accompanied by an enhanced thermal front, whereas the subtropical region is characterized by a weaker westward flow in the surface layer and a thermocline uplifted in the southern part of the domain.

2.2. Eddy Identification, Categorization and Compositing Algorithm

The eddy identification and tracking method we employed in this study is the py-eddy-tracker (PET) algorithm developed by Mason et al. (2014), which is based on Chelton et al. (2011). The PET method has been widely applied to global and regional eddy identification (Pegliasco et al., 2022; Umbert et al., 2015). Eddy identification is performed by scanning closed SSH contours at 0.2 cm intervals. The center of the closed SSH contour within the eddy is defined as the location of the eddy. The eddy-tracking algorithm is then applied to identify eddy trajectories lasting over 10 days, allowing for up to 3 days of missing detections to maintain the continuity of tracking. The PET algorithm outputs provide information on eddy properties, including position, radius, amplitude, and other characteristics. By applying this method to the 1-year simulation, a total of 514 (514) eddy trajectories with 17029 (13725) snapshots of cyclonic (anticyclonic) eddies are found in the subtropical region, and a total of 454 (601) eddy trajectories with 12159 (14571) snapshots of cyclonic (anticyclonic) eddies are

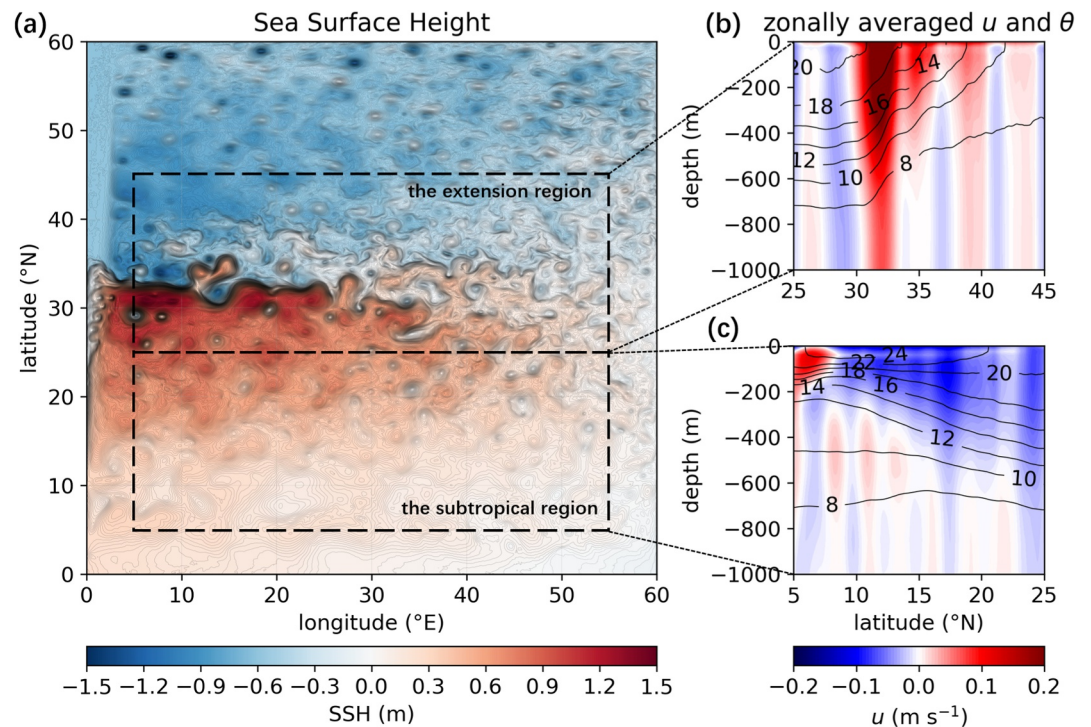


Figure 1. (a) Horizontal distribution of SSH (m) in the domain, with dashed boxes indicating the extension and subtropical regions. (b, c) Vertical distribution of zonally averaged u (colors, m s^{-1}) and temperature (θ , contours, $^{\circ}\text{C}$) in the upper 1,000 m in (b) the extension region and (c) the subtropical region.

found in the extension region. To quantitatively characterize the multipole structure of vertical velocity within mesoscale eddies, we develop a simple algorithm for eddy categorization based on a polar coordinate framework by employing an azimuthal decomposition of the vertical velocity field at the eddy radius (Figure 2). Specifically, 24 desired radii are uniformly chosen around the eddy center in each snapshot, along which the vertical velocity is radially integrated. The resulting values are then arranged sequentially in a counterclockwise direction to construct a continuous distribution of vertical velocity as a function of azimuthal angle. For example, the monopole structure is classified by the signal of integrated vertical velocity maintains a nearly constant sign throughout the 360° rotation, or the mean value significantly exceeds the amplitude of any oscillating components (Figure 2a). In this mode, the number of zero-crossings is typically zero or one (Figure 2b). Similarly, the dipole and quadrupole structures have 2 and 4 zero-crossings across the 360° phase, respectively. By analyzing the variation in phase and amplitude along the curve, we reveal the spatial distribution of the multipole structure with alternating positive and negative extrema, and classified the different types of multipole vertical velocity observed within individual eddy snapshots.

To capture the fundamental features of the eddy-induced dipole vertical velocity structures within complex background dynamic environments, we conduct composite analyses for dipole cyclonic and anticyclonic eddies in both the subtropical and extension regions. For any desired variable φ (e.g., SSH, vertical velocity), we can match eddy observations from the PET algorithm with φ in time and space. These collocated observations are then binned in the “monopole”, “dipole”, and “quadrupole” structures respectively after executing the above-mentioned categorization algorithm to produce composites of φ for different structures. Specifically, for each sub-category of eddies identified by the classification algorithm (e.g., dipole cyclonic eddies in the subtropical region), we first defined the eddy center—determined by the PET algorithm—as the origin of the coordinate system. All eddy snapshots were normalized by their respective eddy radius (R) and subsequently interpolated onto a uniform Cartesian grid extending ± 2 normalized radius, with a grid resolution of $\Delta x = \Delta y = 0.04$. Within this normalized domain, the vertical velocity fields from all identified snapshots were averaged at each corresponding depth layer. Notably, we deliberately avoided additional coordinate alignment or rotation (e.g., rotating the snapshots according to sign of dipole or the direction of eddy propagation) to prevent imposing artificial

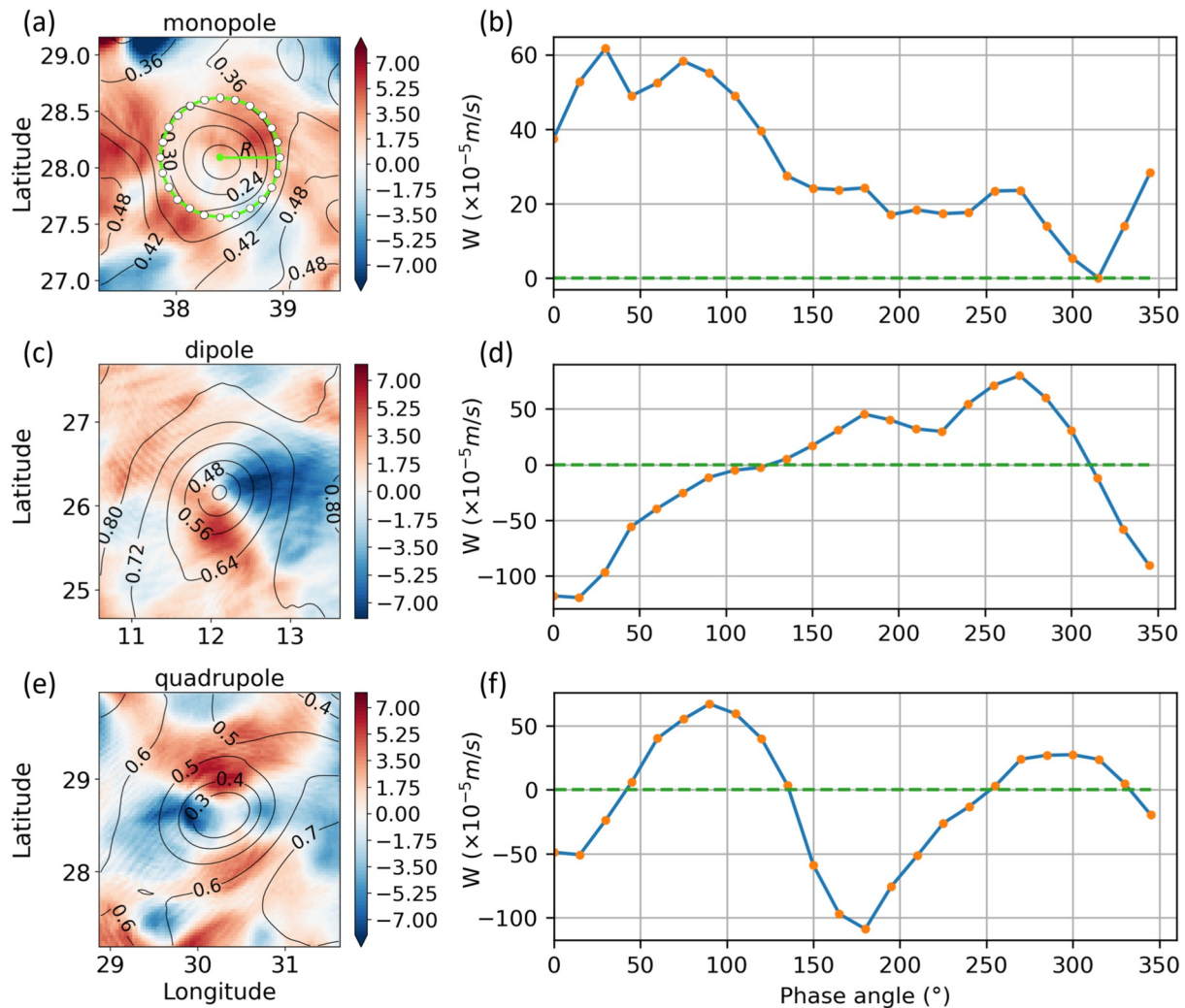


Figure 2. Horizontal distribution of (a) monopole, (c) dipole, and (e) quadrupole vertical velocity (colors, m s^{-1}) identified by the categorization algorithm, taking cyclonic eddies in the extension region as examples. The light green dot and solid line denote the eddy center and radius, respectively. The white dots represent the position of the 24 radii on the eddy circle and the black solid contours denote the SSH (m). (b) The variation of integrated vertical velocity along the azimuthal angle at eddy radius (R) in the monopole structure. The blue line with orange dots represents the integrated vertical velocity along azimuthal angle and the green dashed line denotes the zero line. Panels (d, f) are same as (b) but for dipole and quadrupole structures, respectively.

constraints or biases on the intrinsic eddy structures. Besides, while the above procedure is repeated at each available vertical level in the upper 1,000 m, we focus on depths of 450 m in Section 3 as it exhibits a stronger vertical velocity throughout the water column.

3. Results

3.1. General Features of Mesoscale Eddies and Associated Vertical Motions Within the Numerical Model

To demonstrate that our model configuration accurately reproduces the fundamental characteristics of mesoscale eddies, we first evaluate the amplitude, radius, and lifetime of cyclonic eddies and anticyclonic eddies in both the subtropical and extension regions (Figure 3). The simulated eddies exhibit characteristic scales and amplitudes that are consistent with global satellite altimetry observations (e.g., Chelton et al., 2011). In the subtropical region, the mean amplitude is approximately 0.076 m (CE) and 0.036 m (AE) (Figure 3a). In the extension region, where the background flow is more energetic, the eddies are more intense, with mean amplitudes reaching 0.209 m (CE) and 0.134 m (AE) (Figure 3d). These results align with established global statistics showing that large-amplitude eddies are predominantly concentrated in regions of highly unstable currents (e.g., extension regions), while mean

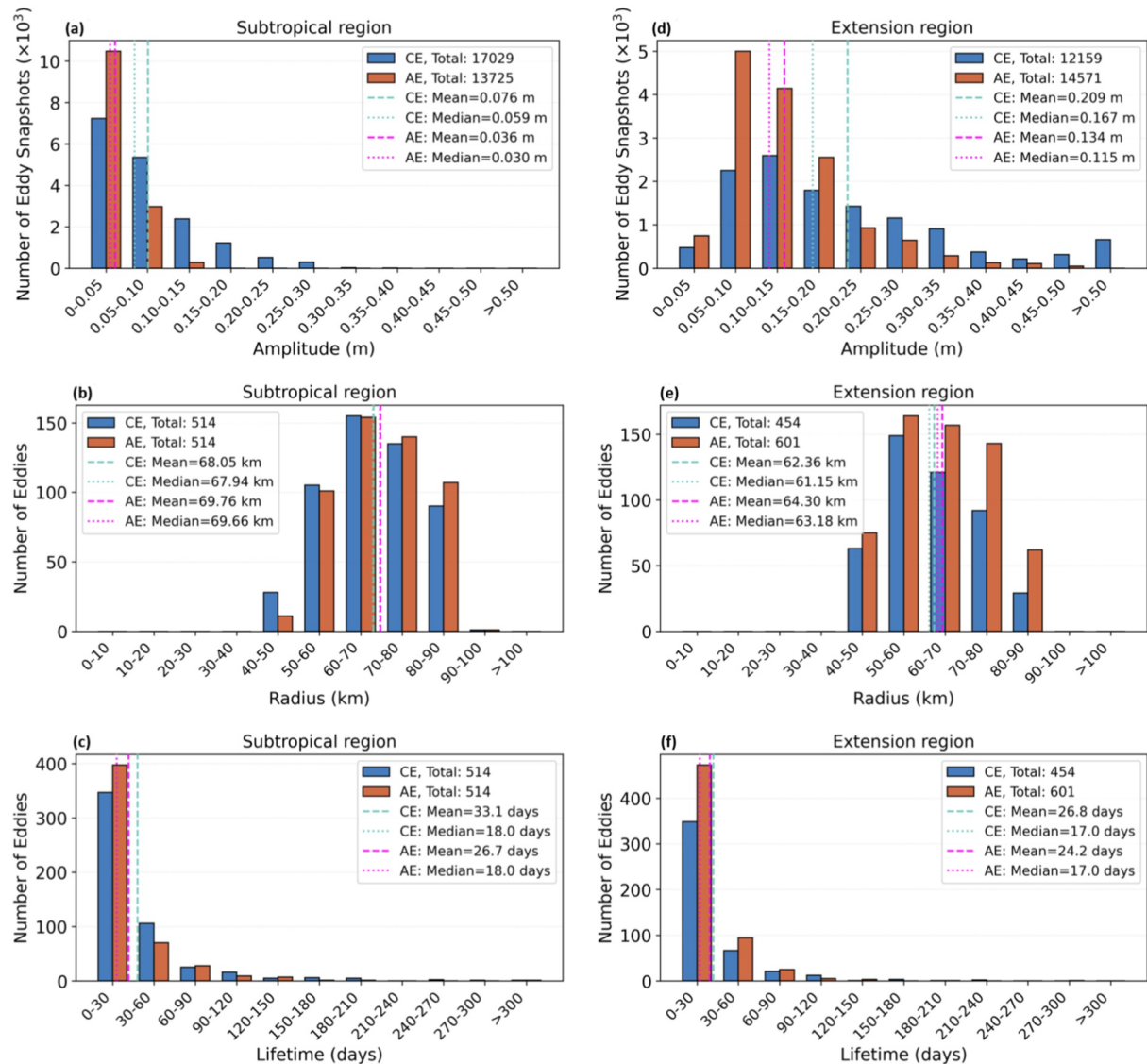


Figure 3. Statistics of the (a) amplitude, (b) radius, and (c) lifetime of cyclonic eddies (CE, blue) and anticyclonic eddies (AE, orange) in the Subtropical region. The light blue dashed line and the magenta dashed line denote the mean and median values of the cyclonic eddies and anticyclonic eddies, respectively. Panels (d–f) are same as (a–c) but for extension region.

amplitudes in subtropical gyres are typically less than 0.1 m (Chelton et al., 2011). Despite the regional discrepancies in amplitude, the radius distributions remain similar between the two regions (Figures 3b and 3e). The radii of tracked eddies predominantly fall between 50 and 100 km, with a characteristic mean scale of 60–70 km, which is consistent with the results of Chelton et al. (2011).

Furthermore, the results indicate that the average lifetime of mesoscale eddies in the model is approximately 4 weeks, with the eddy counts decreasing rapidly as lifetime increases (Figures 3c and 3f). Specifically, eddies in the subtropical region exhibit slightly longer lifetime (mean of 33.1 days for CE and 26.7 days for AE) compared to those in the extension region (26.8 days for CE and 24.2 days for AE). This regional discrepancy is likely attributable to the enhanced instability and energy dissipation processes induced by the complex background dynamics and intense eddy-mean flow interactions in the extension region. While the overall eddy lifetime in our simulation is relatively short, this characteristic does not compromise our subsequent analysis, as the primary objective of this work is to investigate the fundamental structure and mechanisms of eddy-induced vertical motion rather than its temporal evolution throughout the complete eddy lifecycle.

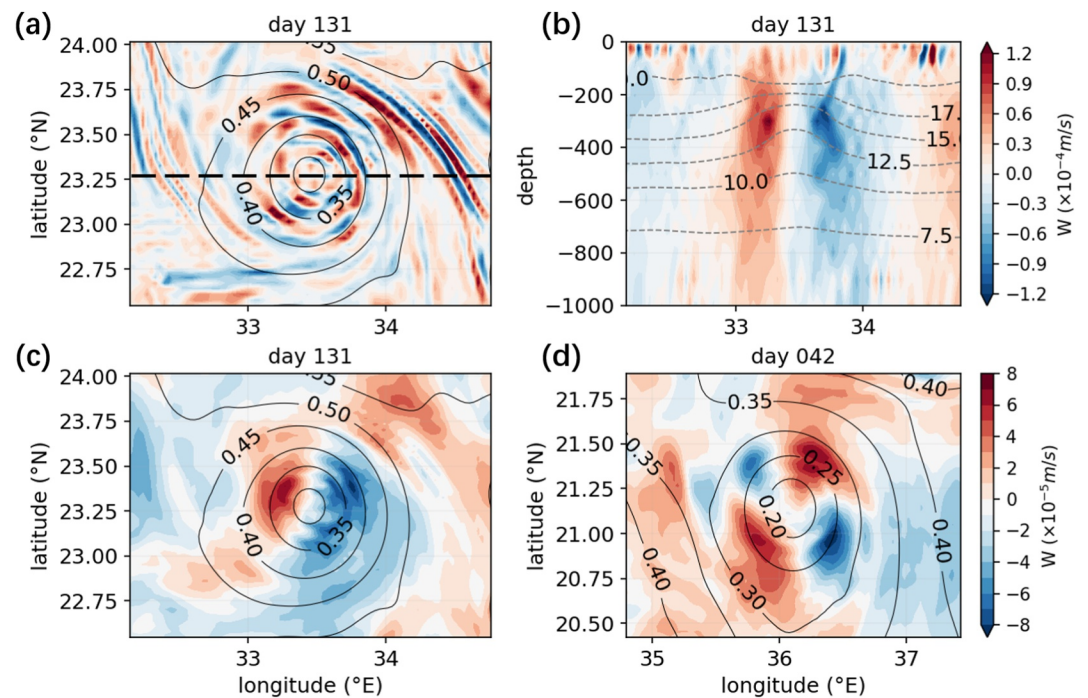


Figure 4. Vertical velocity (m s^{-1}) within an individual cyclonic eddy for (a) at 50 m on day 131, (b) the zonal section at the eddy center on day 131, and the vertical velocities averaged from 100 to 500 m on (c) day 131 and (d) day 042. The solid contour denotes the SSH (m) while the gray dashed line denotes the temperature ($^{\circ}\text{C}$).

In conclusion, while the relatively idealized forcing is set in the numerical model, the mesoscale eddies are generated through intrinsic instabilities processes, which ensures that the simulated eddies capture the essential nonlinear dynamics and three-dimensional structures of that in real-ocean, thereby providing a reliable foundation for our further diagnostic analysis.

To obtain the general features of the vertical velocity structures, we examine the instantaneous vertical velocity fields of simulated eddies. The results reveal multipole structures within mesoscale eddies, including dipole and quadrupole patterns, as well as the well-known monopole pattern. With a horizontal resolution of approximately 3 km, our model can adequately resolve mesoscale processes and partially capture submesoscale dynamics, enabling a detailed examination of their vertical velocity structures.

Figure 4 shows the horizontal distribution of vertical velocity within an individual cyclonic eddy from our numerical simulation. The vertical velocity in the surface layer exhibits complex and ubiquitous filaments characterized by widths of approximately 10 km and depths of up to 100 m (Figures 4a and 4b), suggesting the presence of substantial submesoscale features (Klein & Lapeyre, 2009). These filaments primarily occur in the eddy periphery and are accompanied by alternating strong upwelling and downwelling motions, which is consistent with the intense transport of heat and biochemical elements in previous studies (Su et al., 2018; Zhang & Qiu, 2020). However, strong vertical velocities associated with submesoscale filaments are confined to the mixed layer (Figure 4b). In contrast, vertical velocities in the deeper ocean display distinct larger scale multipole structures, such as dipole and quadrupole patterns (Figures 4c and 4d), with the maximum values occurring at a depth of 300–400 m (Figure 4b). The dipole structure consists of paired upwelling and downwelling cells separated by the eddy center, whereas the quadrupole configuration exhibits radially distributed alternating upwelling and downwelling cells around the eddy center. Notably, vertical velocities are relatively stronger at the eddy periphery than in the interior.

To quantify the prevalence of different vertical velocity structures, we employ the eddy categorization algorithm to analyze snapshots within the subtropical and extension regions. The results shown in Figure 5 demonstrate that nearly all snapshots can be categorized into one of three types: monopole, dipole, or quadrupole. Notably, dipole structures emerge as the dominant pattern in both regions, accounting for 67.9% (61.6%) of cyclonic

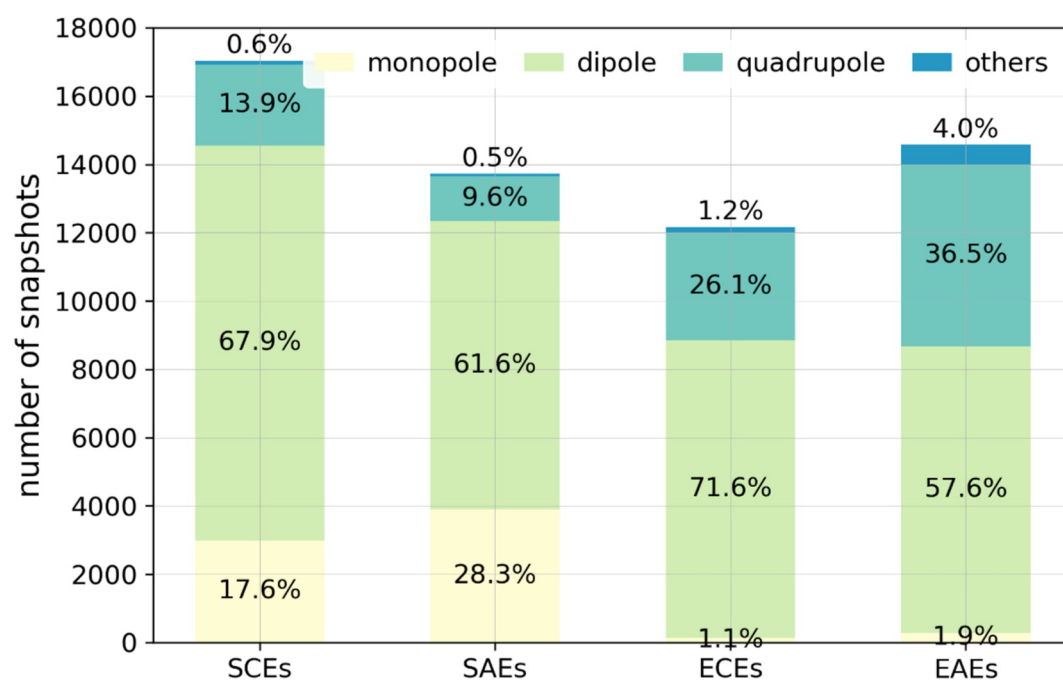


Figure 5. Percentage of multipole structures for Subtropical Cyclonic Eddies (SCEs), Subtropical Anticyclonic Eddies (SAEs), Extension Cyclonic Eddies (ECEs), and Extension Anticyclonic Eddies (EAEs). The light yellow, green, cyan, and blue colors indicate monopole, dipole, quadrupole, and other structures, respectively.

(anticyclonic) eddies in the subtropical region and 71.6% (57.6%) of cyclonic (anticyclonic) eddies in the extension region. Furthermore, although the monopole structures are not dominant in either region, they are more common in the subtropical region compared with the extension region, accounting for 17.6% (28.3%) of cyclonic (anticyclonic) eddies. Quadrupole structures occur more frequently in the extension region than in the subtropical region, accounting for 26.1% (36.5%) of cyclonic (anticyclonic) eddies. Beyond the three dominant multipole structures, other structures (e.g., octupole patterns, categorized as 'Others' in Figure 5) occur only rarely and account for a negligible proportion in all cases.

3.2. Composite Analysis of the Dipole Structure

To investigate the general characteristics of the dipole structure, we conduct a composite analysis on dipole patterns within the Subtropical Cyclonic Eddies (SCE), the Subtropical Anticyclonic Eddies (SAE), the Extension Cyclonic Eddies (ECE), and the Extension Anticyclonic Eddies (EAE). In the subtropical region, the composite results reveal zonal dipole patterns with alternating upwelling and downwelling cells separated by the eddy center. The strongest vertical motions occur at the peripheries of the eddies and can affect twice the eddy radius (Figures 6a and 6b). In the vertical dimension, the maximum vertical velocities for both SCE and SAE are observed at 300–400 m, with their vertical influence persisting to 1,000 m (Figures 6c and 6d). Notably, the dipole structures of SCE and SAE exhibit opposite patterns: upwelling (downwelling) occurs to the west (east) of the SCE, whereas the SAE shows upwelling (downwelling) to the east (west).

A composite analysis in the extension region reveals dipole structures that are opposite to those of the subtropical region, both in the cyclonic eddy and the anticyclonic eddy. Specifically, upwelling (downwelling) cell occurs to the east (west) of the ECE, while the EAE shows upwelling (downwelling) to the west (east). Besides, the maximum vertical velocities for both ECE and EAE are observed at a depth of about 400 m (Figures 7c and 7d). Notably, stronger vertical velocities are observed in the extension region, which are associated with higher amplitude.

While similar structures have been documented in early studies, the regional contrasts in dipole patterns receive insufficient attention (Estrada-Allis et al., 2019; Mason et al., 2017). These contrasting dipole patterns between

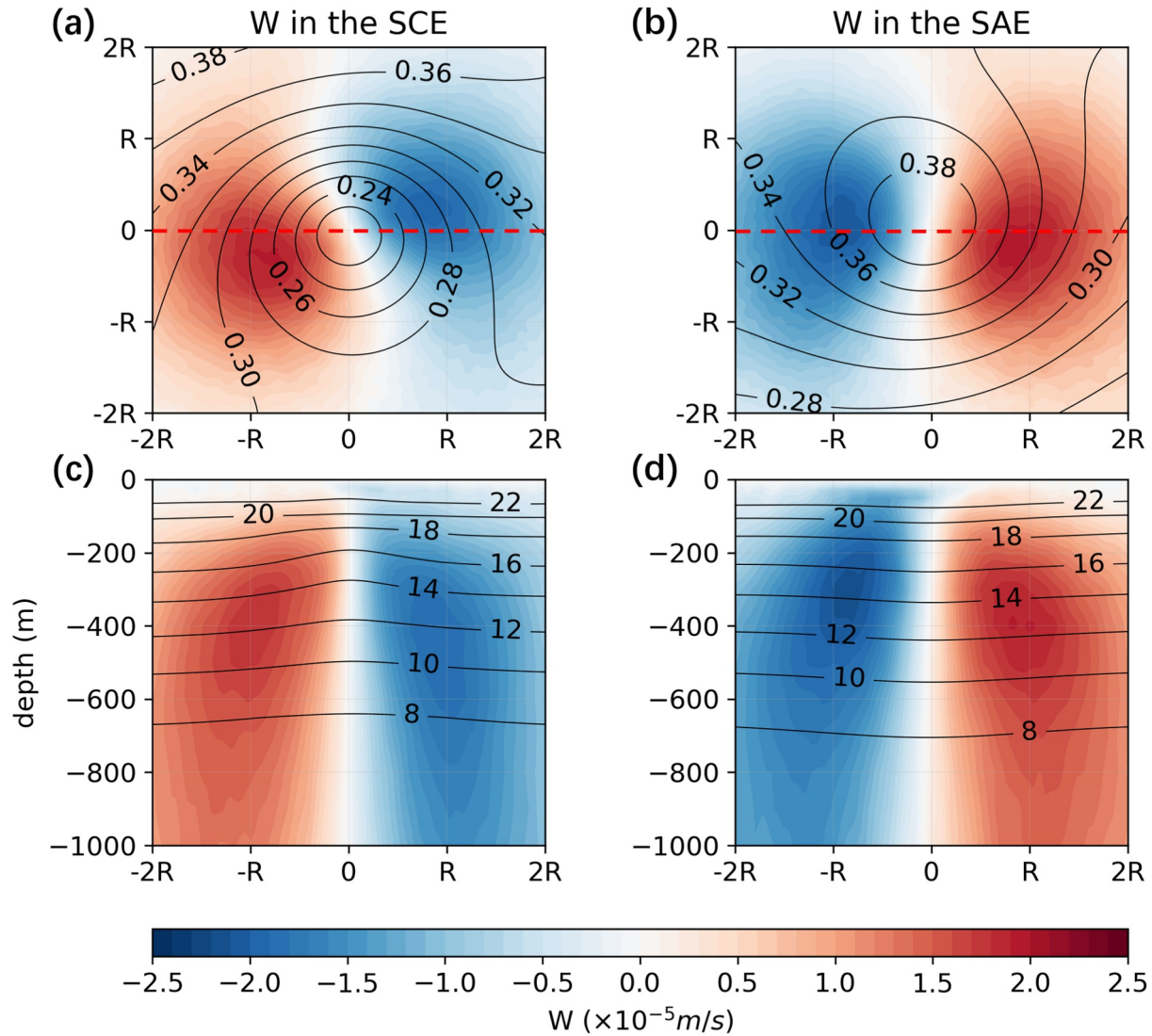


Figure 6. (a) The horizontal distribution and (c) zonal section of the composited vertical velocity (W , colors, m s^{-1}) within the SCE. The horizontal SSH (m) and vertical temperature ($^{\circ}\text{C}$) are represented by black contours in (a, c) respectively. The red dashed line represents the zonal section across the eddy center. Panels (b, d) are the same as (a, c) but for SAE. The zonal and meridional coordinates of the composite averages are normalized ranging from -2 to $2R$ (R is the eddy radius).

the two regions may arise from the influence of the distinct background larger-scale circulation pattern, underscoring the need for further investigation into the underlying physical mechanisms across the two regions.

3.3. Diagnosis With the Vorticity Equation

Here, we diagnose the vorticity equation to decompose the vertical velocity field into components with various forcing terms. This approach expresses vertical velocity using the horizontal velocity components by considering continuity (Koszalka et al., 2009). The detailed decomposition and derivation of the diagnosed equation are as follows, with the initial momentum conservation equation in the MITgcm given by:

$$\frac{\partial \mathbf{V}}{\partial t} = -\mathbf{V} \cdot \nabla \mathbf{V} - 2\boldsymbol{\Omega} \times \mathbf{V} - \frac{1}{\rho} \nabla P + \mathbf{F}_{\text{Ext}}, \quad (2)$$

where $\mathbf{V}(u, v)$ is the horizontal velocity vector, $\nabla = \frac{\partial}{\partial x} \mathbf{i} + \frac{\partial}{\partial y} \mathbf{j} + \frac{\partial}{\partial z} \mathbf{k}$ is the Nabla operator, $\boldsymbol{\Omega}$ is the rotating angular speed of the Earth, ρ is the water reference density, and P is the pressure. In Equation 2, $\frac{\partial \mathbf{V}}{\partial t}$ is the tendency

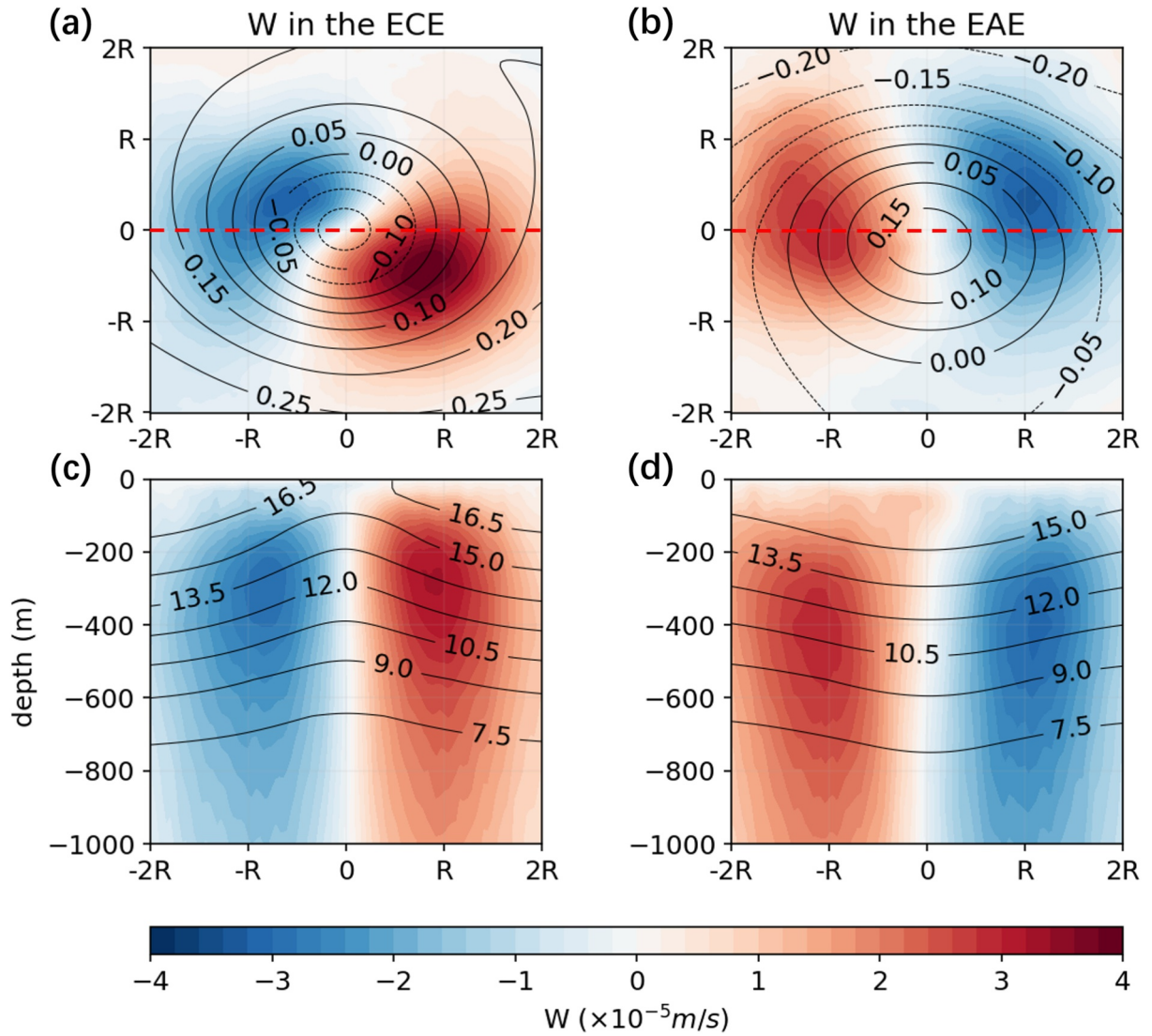


Figure 7. The same as Figure 6 but for the ECE and EAE composites.

of the velocity field. $-V \cdot \nabla V$ is the advection term of velocity, $-2\Omega \times V$ is the Coriolis term, $-\frac{1}{\rho}\nabla P$ is the pressure gradient, and F_{Ext} represents friction, dissipation and wind forcing terms. We then calculate the curl of the horizontal momentum equations through cross-differentiation by $-\frac{\partial}{\partial y}$ and $\frac{\partial}{\partial x}$, which gives:

$$\frac{\partial}{\partial t} \left(-\frac{\partial u}{\partial y} \right) = \frac{\partial}{\partial y} (V \cdot \nabla u) - \beta v - f \frac{\partial v}{\partial y} + \frac{1}{\rho} \frac{\partial}{\partial y} \frac{\partial P}{\partial x} - \frac{\partial}{\partial y} F_{\text{Ext},x} \quad (3)$$

and:

$$\frac{\partial}{\partial t} \left(\frac{\partial v}{\partial x} \right) = \frac{\partial}{\partial x} (-V \cdot \nabla v) - f \frac{\partial u}{\partial x} - \frac{1}{\rho} \frac{\partial}{\partial x} \frac{\partial P}{\partial y} + \frac{\partial}{\partial x} F_{\text{Ext},y} \quad (4)$$

Combining Equations 3 and 4 yields:

$$(f + \zeta) \frac{\partial w}{\partial z} = \frac{\partial \zeta}{\partial t} + V \cdot \nabla \zeta + \frac{\partial w}{\partial x} \frac{\partial v}{\partial z} - \frac{\partial w}{\partial y} \frac{\partial u}{\partial z} + \beta v - V_{\text{orExt}} \quad (5)$$

where $\zeta = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}$ is the vertical component of the relative vorticity, and the term $\left(\frac{\partial w}{\partial x} \frac{\partial v}{\partial z} - \frac{\partial w}{\partial y} \frac{\partial u}{\partial z}\right)$ is the tilting effect of ζ . To simplify the calculations, both $\zeta \frac{\partial w}{\partial z}$ and $\left(\frac{\partial w}{\partial x} \frac{\partial v}{\partial z} - \frac{\partial w}{\partial y} \frac{\partial u}{\partial z}\right)$ are omitted because they are approximately two orders of magnitude smaller than the other terms. Vor_{Ext} represents dissipation and other external forcings derived from $\mathbf{F}_{\text{Ext}}$. Further integration of Equation 5 yields the following diagnostic equation:

$$w(x, y, z) = \frac{1}{f} \left(\int_z^\eta \frac{\partial \zeta}{\partial t} dz + \int_z^\eta \mathbf{V} \cdot \nabla \zeta dz + \int_z^\eta \beta v dz - \int_z^\eta \text{Vor}_{\text{Ext}} dz \right) \quad (6)$$

The first term on the right-hand side (rhs) formulates the contribution of the instantaneous change of the ζ (hereafter, w_{Tend}), while the second term on the rhs represents the contribution of advection (w_{Adv}). $w_{\text{Ageo}} = w_{\text{Tend}} + w_{\text{Adv}}$ is the sum of the above two terms, representing the material derivative of vorticity. In mesoscale and submesoscale processes that departure from geostrophic balance, this material derivative directly corresponds to the imbalance of vorticity (Chen et al., 2020; Koszalka et al., 2009). Consequently, this combined term captures the ageostrophic effect required to maintain dynamical consistency, serving as a primary indicator of “unbalanced” motion in vorticity diagnostics. The third term is derived from the Coriolis term in Equation 2 and represents the contribution of β -effect to the vertical velocity, and hereafter referred to as w_{Beta} . This term will enable the evaluation of the individual contribution of the β -effect. The final rhs term in Equation 6 accounts for the influence of external forcing, like the interactions of the eddy with the background current, other eddies, bathymetry, and so on (hereafter, w_{Ext}). Note that although the momentum terms employed in the derivation are based on daily-averaged outputs, the resulting balance of Equation 6 is found to be nearly conserved within the depth of interest. Based on the preceding composite analysis of the vertical distribution of dipole structures, the strongest vertical motions typically occur at a depth of approximately 400–500 m (Figures 6c, 6d and 7c, 7d). Thus, a depth of $z = 450$ m is selected for integration and diagnostic evaluation.

The dynamics associated with the individual forcing terms in the subtropical region are shown in Figure 8. The diagnosed vertical velocity (black dash dotted line) shows a consistent order-of-magnitude on 10^{-5} m s^{-1} with the composite results of direct model outputs (Figures 4 and 5). Note that the y-axis in our plot is scaled to 10^{-4} to better resolve the relative contributions of individual terms across the eddy radius. For both cyclonic and anticyclonic eddies, the normalized results reveal that the terms w_{Tend} and w_{Adv} make dominant contributions within the eddy radius (Figures 8a and 8c). The w_{Beta} acts as a secondary forcing and w_{Ext} (including effects of friction, dissipation and wind forcing) is approximately one order of magnitude smaller than the dominant terms in a depth of 450 m (Figures 8a and 8c). We note that the contributions from w_{Tend} and w_{Adv} largely cancel each other out (Figures 8b and 8d), with the sum of these two terms (w_{Ageo}) represents the contribution of ageostrophic processes to the vorticity and can also be interpreted as the distance from geostrophic balance (Koszalka et al., 2009; Zhong et al., 2012). Due to weak background flow in the subtropical region, the tendency and advection of vorticity are generally in balance within the eddy, indicating a relatively weak ageostrophic effect and making the w_{Ageo} a secondary contribution. Therefore, in Figure 8, although w_{Ageo} and w_{Beta} showing similar contributions from core to periphery, the vertical velocity induced by the β -effect dominates the dipole structures.

In the extension region, the terms w_{Tend} , w_{Adv} , and w_{Beta} remain as the primary contributors to the dipole pattern, as shown in Figure 9. However, these terms exhibit a more complex and asymmetric pattern compared to the subtropical region due to the influence of the strong background current. Intense w_{Ageo} arises in locations where w_{Tend} and w_{Adv} do not cancel each other out, causing a strong vorticity imbalance that drives the vertical velocity. Thus, the term w_{Ageo} , which is balanced by the term w_{Beta} , further dominates the dipole pattern in this region. Therefore, the dipole pattern of eddy vertical velocity is fundamentally controlled by the relative strength of the ageostrophic effect and the β -effect.

3.4. Diagnosis With the QG-Omega Equation

The previous diagnostic analysis reveals the important contributions of the β -effect and the ageostrophic effect, which provides a direct interpretation of the dynamical balance between individual terms. However, the physical processes underlying the zonal opposite dipoles across the two regions and eddy polarities remains unclear. The advection and tendency terms of the vorticity exhibit great differences across the two regions, indicating that the distinct background fields may play an important role. At the depth we focused, the dynamics of mesoscale eddies

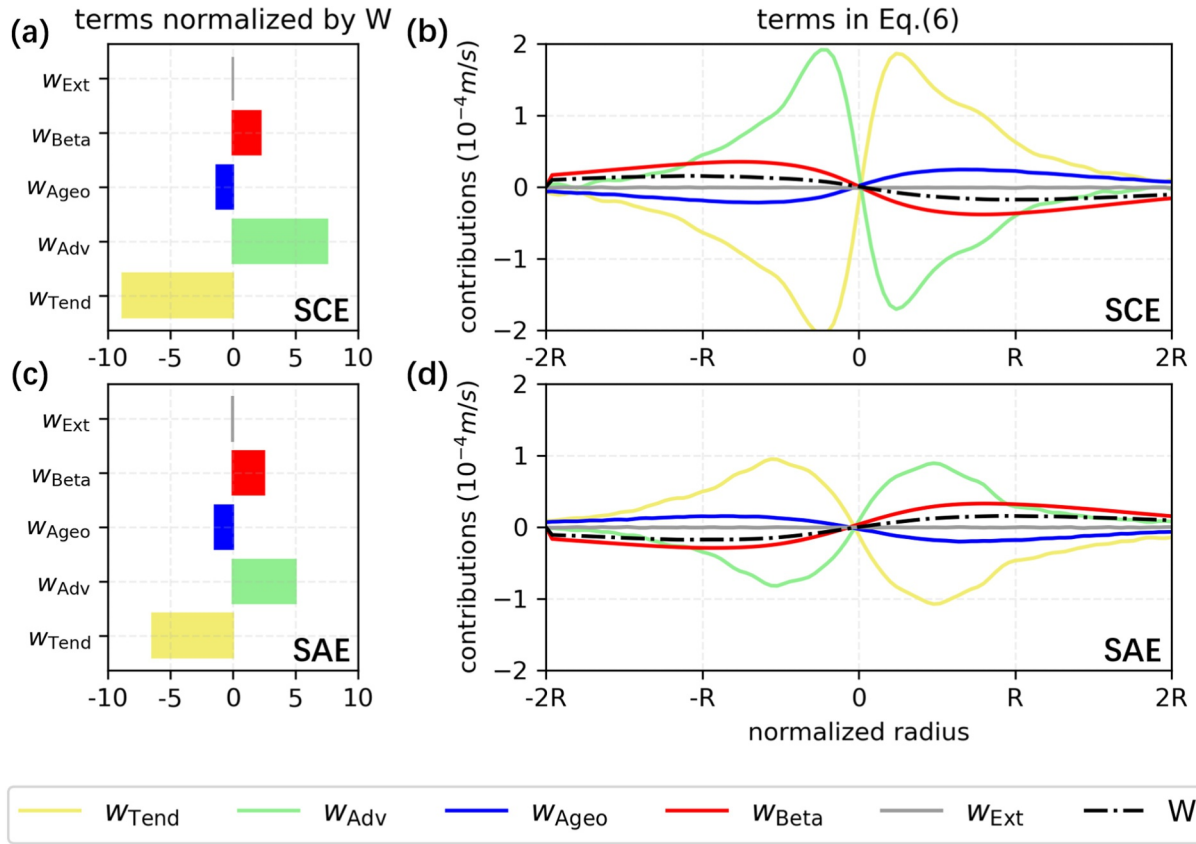


Figure 8. (a) Contributions of the forcing terms in Equation 6 for the SCE normalized by the vertical velocity (W) at a depth of 450 m. (b) Contributions of each forcing term in Equation 6 to the vertical velocity for the SCE, with $w_{\text{Ageo}} = w_{\text{Tend}} + w_{\text{Adv}}$. The x-coordinate is normalized radius of the composite result in zonal direction. Panels (c, d) are the same as (a, b) but for the SAE.

are governed primarily by quasi-geostrophic balance. Within this framework, an alternative physical interpretation of the vertical velocity can be obtained from the QG omega equation, which has been applied to diagnose mesoscale vertical velocity in the ocean (e.g., Ni et al., 2020; Pascual et al., 2015). Rather than focusing on individual forcing terms, the QG Omega equation effectively links the vertical velocity field to the kinematic deformation resulting from the interaction between large-scale background currents and density gradients. To quantitatively assess the relationship between dipole structures and the background flow field while comparing their relative importance to the β -effect, we solve the Q-vector form of the QG omega equation on the β -plane, which is:

$$N^2 \nabla_h^2 w + f^2 \frac{\partial^2 w}{\partial z^2} = -2 \nabla_h \cdot \mathbf{Q} + f \beta \frac{\partial v_g}{\partial z}. \quad (7)$$

Where N^2 denotes the buoyancy frequency, and f is the Coriolis parameter. The term $\mathbf{Q}$, or Q-vector, is given by:

$$\mathbf{Q} = \frac{g}{\rho_0} \left(\frac{\partial u_g}{\partial x} \frac{\partial \rho}{\partial x} + \frac{\partial v_g}{\partial x} \frac{\partial \rho}{\partial y}, \frac{\partial u_g}{\partial y} \frac{\partial \rho}{\partial x} + \frac{\partial v_g}{\partial y} \frac{\partial \rho}{\partial y} \right). \quad (8)$$

The left-hand side of Equation 7 involves spatial derivatives of vertical velocity. Therefore, the equation can be used to diagnose the field of w in terms of the Q-vector and β -effect on the right-hand side of the equation. The divergence (convergence) of Q-vector corresponds to the downwelling (upwelling) cells of the vertical velocity, which is also exemplified by Calil and Richards (2010). The Q-vector represents the kinematic deformation that arises in shear and confluence situations due to the interactions between the horizontal velocity and the density (Giordani et al., 2006; Pietri et al., 2021). Figure 10 illustrates the horizontal distributions of the vertical velocity

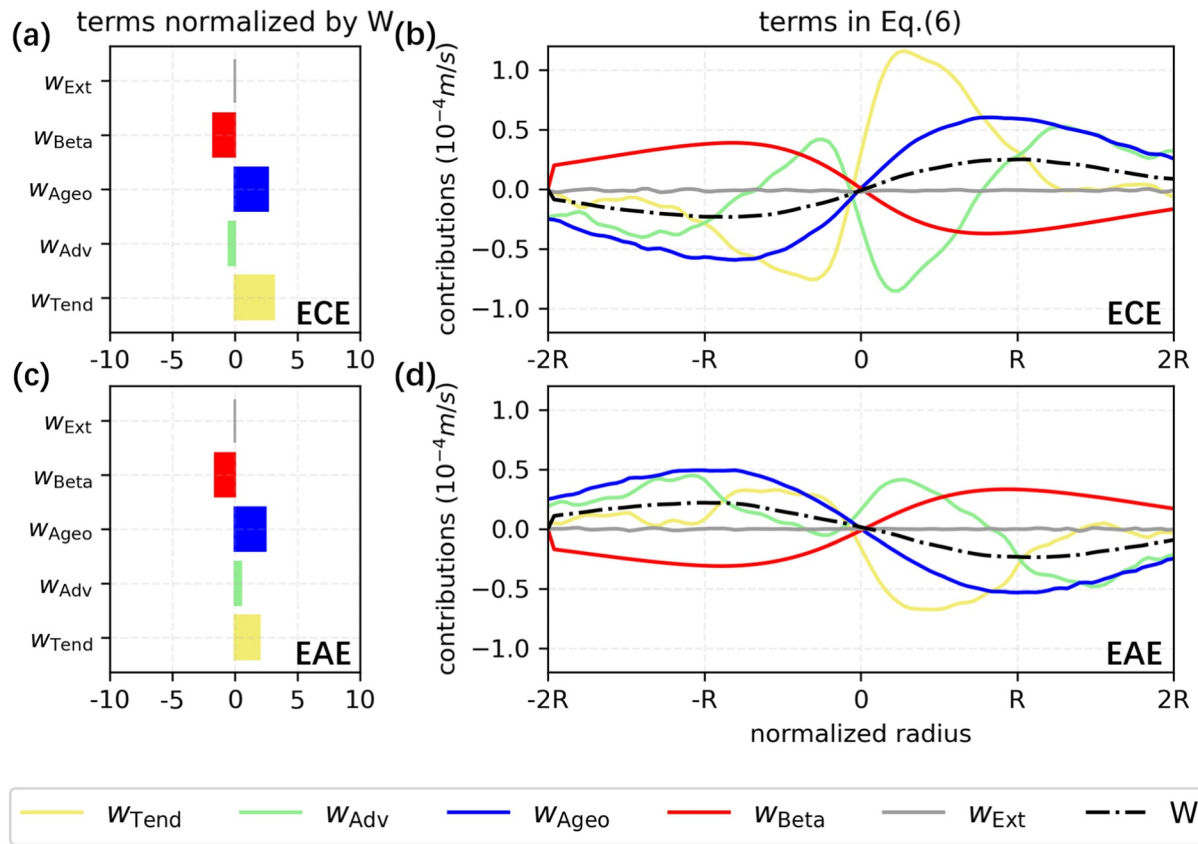


Figure 9. The same as Figure 8 but for the ECE and EAE.

and Q-vector divergence fields for the four composited eddies (SCE, SAE, ECE, and EAE), as diagnosed by the QG Omega equation. The dipole vertical velocity derived from the QG Omega equation exhibit spatial consistency with the composite structures from numerical simulation outputs (Figures 6a, 6b and 7a, 7b) and demonstrate a noteworthy correlation with the Q-vector divergence/convergence patterns. To quantify this consistency, we calculate the spatial correlation between the composited results and the diagnostic vertical velocity fields of the four composite eddies. The spatial correlation coefficients corresponding to the calculations for SCE, SAE, ECE, and EAE reach 0.94, 0.94, 0.91, and 0.85, respectively. We noted that the spatial correlations are higher for the subtropical region (SCE/SAE) compared to the extension region (ECE/EAE) because the fact that eddies in the subtropical region are generally weaker and more stable, whereas those in the extension region are stronger and exhibit more intense variability. Besides, the Q-vector in the SCE converges in the west and diverges in the east, whereas in the SAE, the divergence is prominent in the west while convergence dominates in the east. Opposite patterns are also observed in ECE and EAE, respectively. The results suggest that the difference in the meridional background density gradient associated with the distinct background currents is the key factor driving the opposite oriented dipoles between the two regions.

Building upon the verification that the QG Omega equation effectively diagnoses the spatial distribution of the eddy-induced dipole vertical velocity, we further designed idealized sensitivity experiments to isolate the specific role of the background flow in modulating these structures. Since both the density and velocity fields are superpositions of large-scale background and local eddy components, it is essential to decouple their respective contributions to the vertical motion. By constructing idealized eddy structures and simplified background density field, we can explicitly evaluate the modulation effects of background flows on the dipole structure. Specifically, we first constructed four idealized, circular eddies on a β -plane by applying a polar coordinate projection to the composite potential density and SSH fields of the SCE, SAE, ECE, and EAE. Besides, the representative background density gradients for the subtropical and extension regions were isolated by removing eddy-related anomalies and performing zonal averaging. This process retained only the meridional density gradients, providing

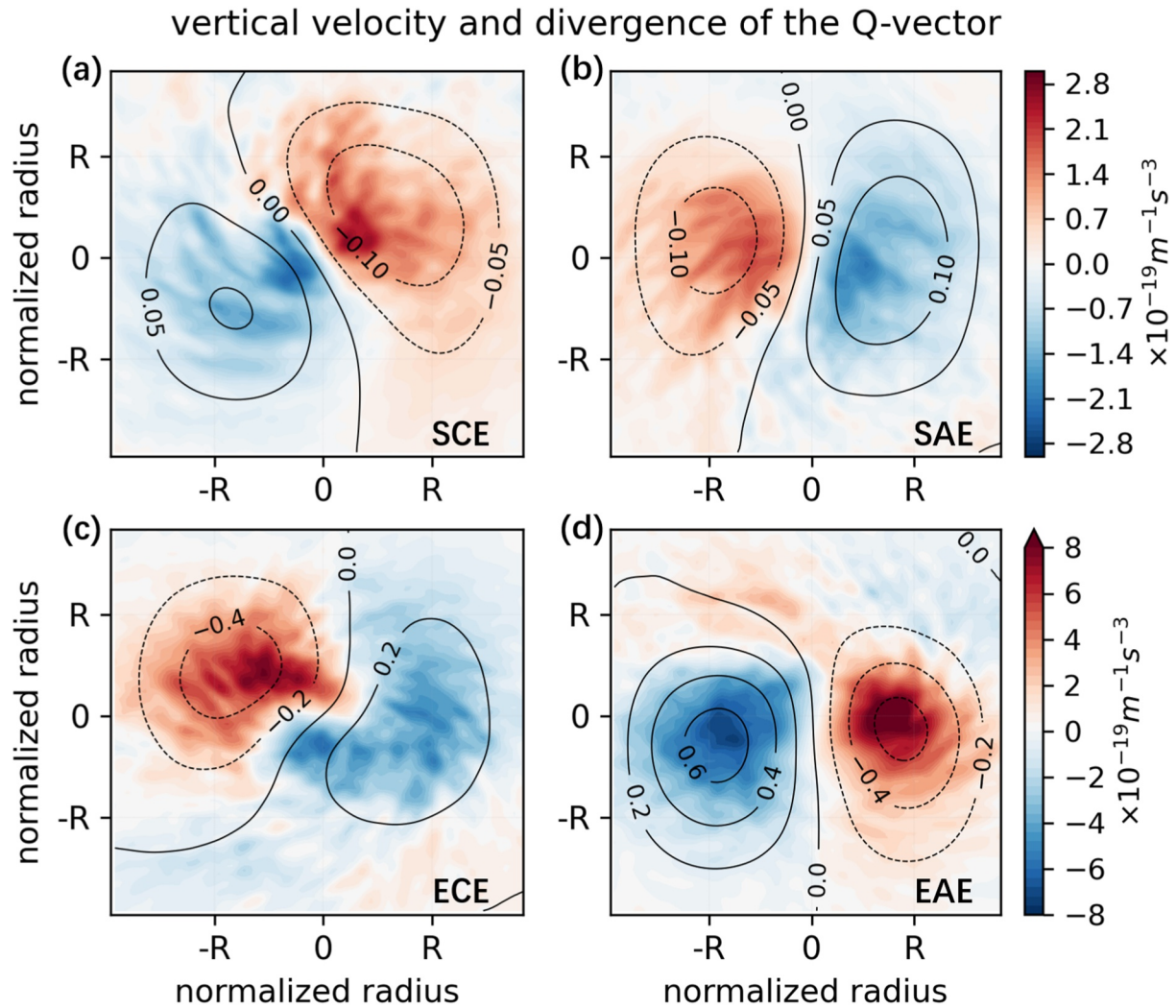


Figure 10. Horizontal distribution of the vertical velocity (contours, $\times 10^{-5} \text{ m s}^{-1}$) and divergence of the Q-vector (colors, $\times 10^{-19} \text{ m}^{-1} \text{ s}^{-3}$) calculated using Equation 7 from composite results (a) SCE, (b) SAE, (c) ECE, and (d) EAE. Red color denotes the divergence and blue color denotes the convergence. Note the different contour intervals between the upper and lower rows.

idealized background states that reflect the distinct dynamical conditions of the targeted regions. The geostrophic velocity fields associated with these idealized eddies and background density states were then calculated separately, enabling a separation of velocity and density variations. The results reveal that because the horizontal geostrophic flow of mesoscale eddies is much stronger than the large-scale background flow, the Q-vector component associated with the local eddy velocity is dominant, while the contribution from the background velocity field is negligible. Consequently, the term $\mathbf{Q}$ in Equation 8 can be decomposed in the Q-vector induced by local eddy density anomaly ($\mathbf{Q}_e$) and that by larger-scale background density gradient ($\mathbf{Q}_b$) as follows:

$$\mathbf{Q}_e = \frac{g}{\rho_0} \left(\frac{\partial u_g}{\partial x} \frac{\partial \rho_e}{\partial x} + \frac{\partial v_g}{\partial x} \frac{\partial \rho_e}{\partial y}, \frac{\partial u_g}{\partial y} \frac{\partial \rho_e}{\partial x} + \frac{\partial v_g}{\partial y} \frac{\partial \rho_e}{\partial y} \right) \quad (9)$$

$$\mathbf{Q}_b = \frac{g}{\rho_0} \left(\frac{\partial u_g}{\partial x} \frac{\partial \rho_b}{\partial x} + \frac{\partial v_g}{\partial x} \frac{\partial \rho_b}{\partial y}, \frac{\partial u_g}{\partial y} \frac{\partial \rho_b}{\partial x} + \frac{\partial v_g}{\partial y} \frac{\partial \rho_b}{\partial y} \right), \quad (10)$$

where $u_g = -\frac{1}{\rho_0 f_0} \frac{\partial p}{\partial y}$ and $v_g = \frac{1}{\rho_0 f_0} \frac{\partial p}{\partial x}$ are the zonal and meridional components of the geostrophic velocity of the above four idealized eddies, respectively; ρ_e is the density anomaly field of the idealized eddies without

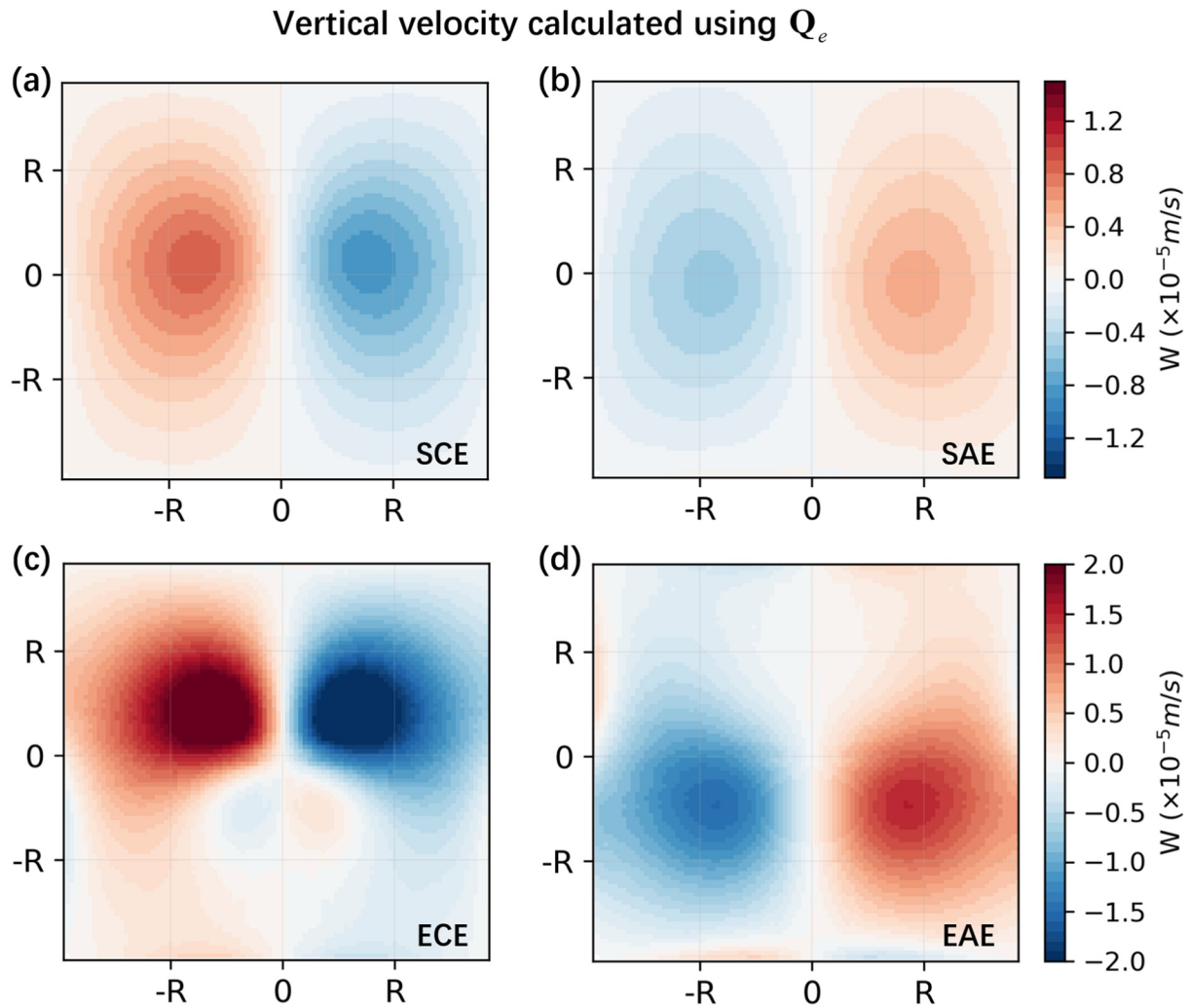


Figure 11. Horizontal distribution of vertical velocity (m s^{-1}) calculated using Q_e instead of Q in Equation 7 for the idealized (a) SCE, (b) SAE, (c) ECE, and (d) EAE.

background density gradient; and ρ_b is the zonally averaged climatological density field associated with the large-scale background, which is characterized by meridional gradients of two regions.

Although the shape of the dipole patterns differs slightly due to the idealized configuration of eddies and background fields, the overall dipole structures are generally aligned with our assumption. Figure 11 shows the vertical velocity calculated with the Q_e , where the general direction of dipole patterns does not present significant variations between the two regions. We notice that the dipole in Figures 11c and 11d exhibit a tendency to transition toward quadrupole configurations, leading to slight differences in their spatial distribution in the extension region. However, the dipole pattern clearly remains the dominant feature. This transition may be associated with instabilities related to stronger eddy intensity in this region, although this is not explored further here. In contrast, calculations using Q_b reveal that eddies of the same polarity display the opposite patterns across the two regions (Figure 12). This indicates that the background density field associated with strong current plays a key role in generating the opposite dipole patterns in the extension region. Specifically, because the background density field has no zonal density gradient, the term involving $\frac{\partial \rho_b}{\partial x}$ vanishes, and the Equation 10 is therefore reduced to the terms containing $\frac{\partial \rho_b}{\partial y}$. Under this condition, if the eddy polarity remains the same (i.e., the velocity-derivative terms in Equation 10 remain unchanged) while the background meridional density gradient has opposite signs in the subtropical and extension regions, the resulting Q_b also changes sign. The differences in the shape of the dipole patterns between the extension and subtropical regions can be attributed to the interaction of the deformation and the β -effect, which would be demonstrated in the following analysis.

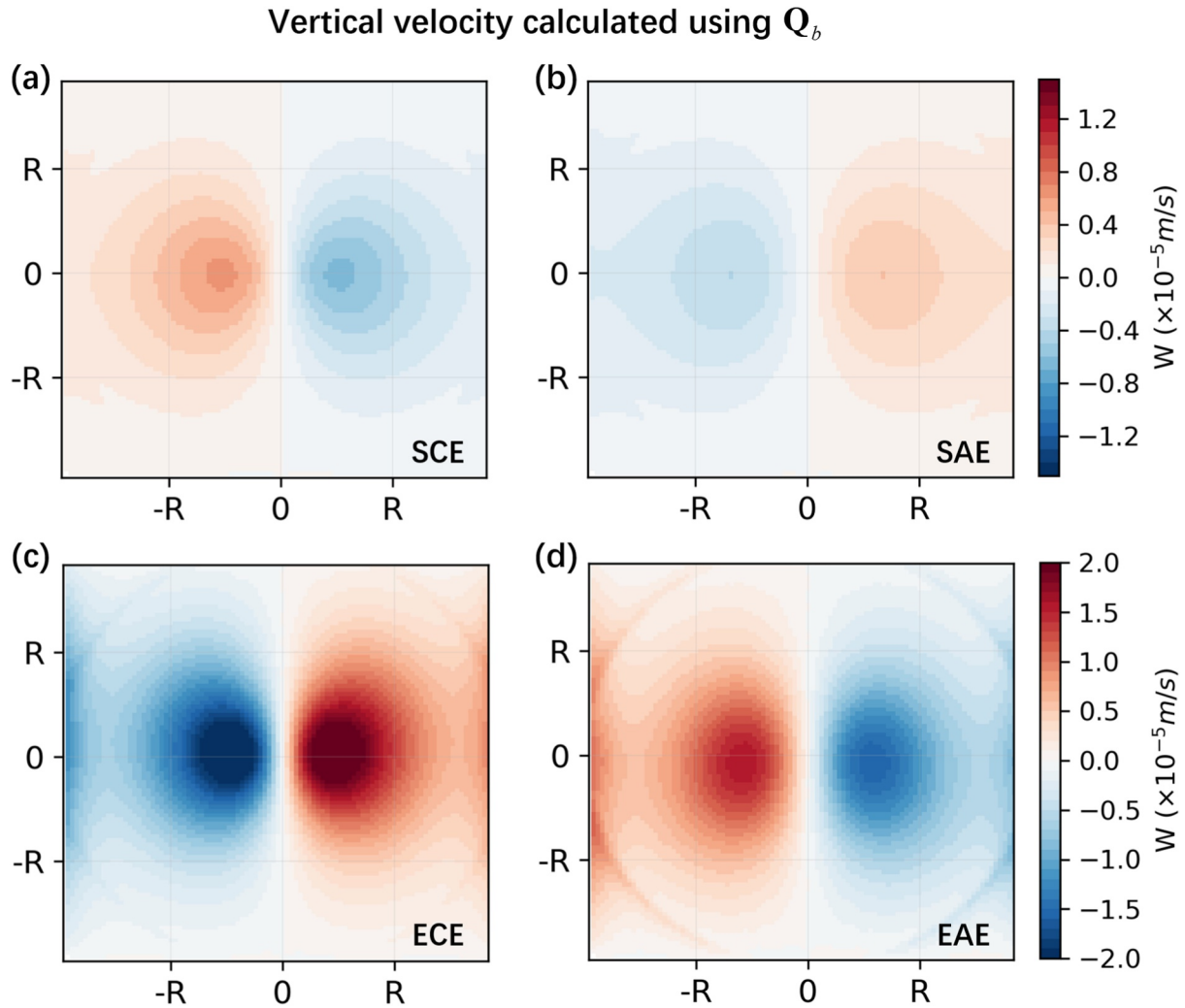


Figure 12. Horizontal distribution of vertical velocity (m s^{-1}) calculated using Q_b instead of Q in Equation 7 for the idealized (a) SCE, (b) SAE, (c) ECE, and (d) EAE.

Since an averaged background state without horizontal density gradient is applied in Equation 9, the corresponding results are expected to represent the basic structure of vertical velocity. Hou et al. (2022) conducted a similar numerical experiment to highlight the role of the β -effect, although their theoretical derivation omitted the Q -vector terms. To verify the influence of deformation effect, we further conduct calculations with the following equation by omitting the Q -vector term from Equation 7:

$$N^2 \nabla_h^2 w + f^2 \frac{\partial^2 w}{\partial z^2} = f \beta \frac{\partial v_g}{\partial z}. \quad (11)$$

The corresponding results for the horizontal distribution of vertical velocity are shown in Figure 13. Vertical velocities induced by the idealized eddies with the same polarity exhibit a consistent spatial pattern in both regions, with upwelling (downwelling) occurring to the west in cyclonic (anticyclonic) eddies. This result aligns well with the idealized experiment on the β -plane conducted by Hou et al. (2022). The observed symmetric spatial distributions in Figure 13 arise as a consequence of the β -effect. Specifically, in the subtropical region, the β -effect aligns with the deformation term in the zonal direction, whereas in the extension region, it acts with an opposite zonal phase relative to the deformation term. This interaction drives the distinct dipole shapes observed within the extension region. As a consequence, the horizontal distribution of the dipole structures in Figure 13 is highly consistent (almost identical) with those shown in Figures 11a, 11b and 12a, 12b, confirming that the influence of the background current can be incorporated by retaining the deformation term in the theoretical

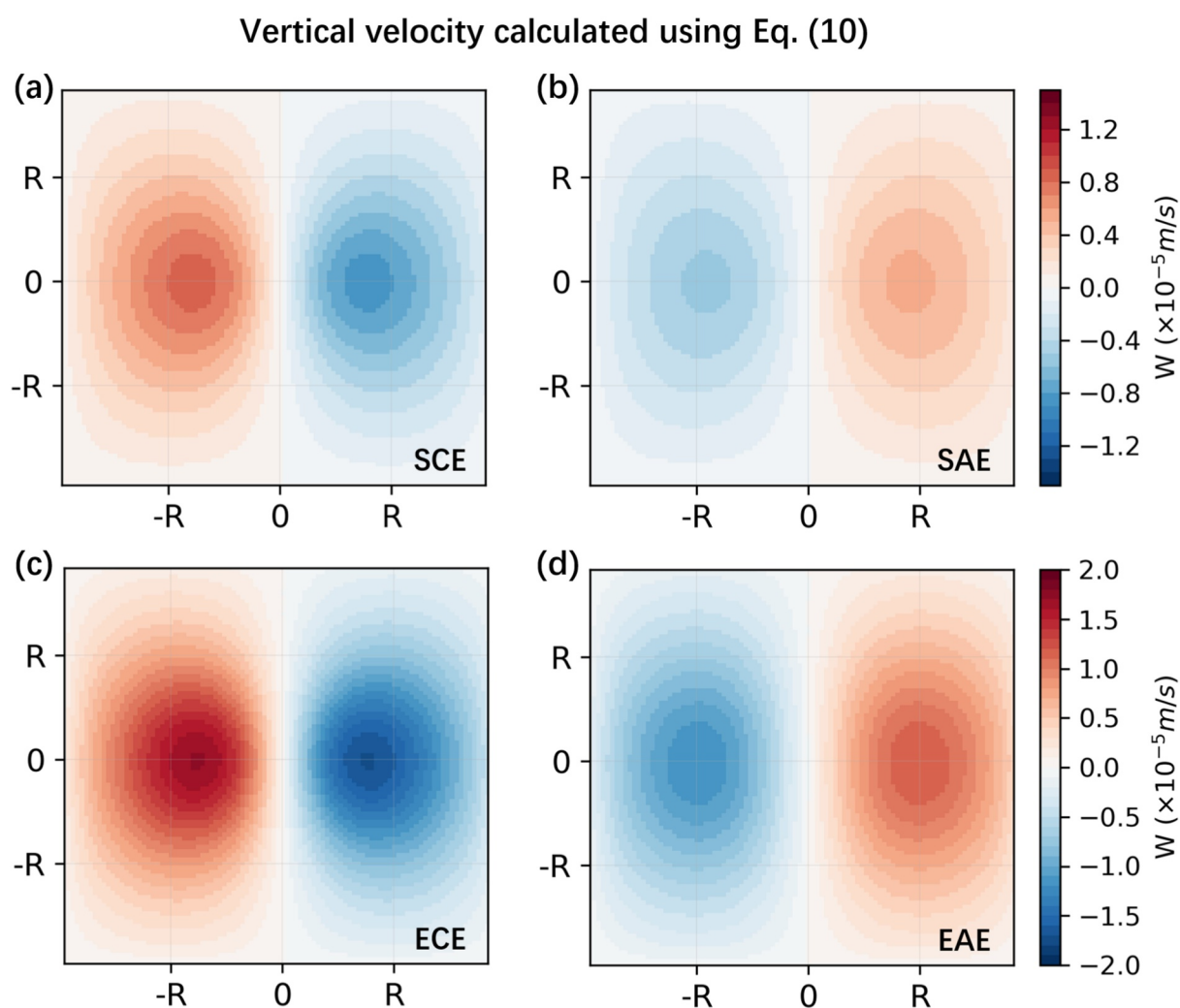


Figure 13. Horizontal distribution of vertical velocity (m s^{-1}) calculated with Equation 11 for the idealized (a) SCE, (b) SAE, (c) ECE, and (d) EAE.

framework. Therefore, our analysis captures the regional variations in dipole structures, which is particularly important in regions characterized by strong background currents.

4. Summary and Discussion

Mesoscale eddies exhibit multipole vertical velocity structures that vary across different hydrodynamic regions. In this study, high-resolution numerical simulation outputs are used to investigate the vertical velocity structures of mesoscale eddies in the subtropical and extension regions. Dipole patterns dominate the vertical velocity structures in both regions, accounting for about 65% of all occurrences, whereas the percentages of monopole and quadrupole structures are much lower. The composite analysis reveals opposite dipole patterns across the different regions and the different eddy polarities. Vorticity diagnostic analysis demonstrates that the dipole structures are governed by the ageostrophic effect and the β -effect. In the subtropical region, the β -effect is largely balanced by the ageostrophic effect in modulating the regional dipole pattern under conditions of a relatively weak background flow. The extension region, in contrast, is strongly influenced by background currents, resulting in the ageostrophic effect becoming the dominant contributor. Moreover, analyses using the QG Omega equation on the β -plane reveal that, in an idealized configuration without background flow, the β -effect modulates the opposite patterns between cyclonic and anticyclonic eddies. In the presence of a strong background flow, however, the deformation effect associated with the eddy-flow interaction induces opposite patterns in the dipole structures between subtropical and extension regions.

Interestingly, the instantaneous pattern of the multipole structures can evolve and rotate during an individual eddy's lifetime. This characteristic is consistent with the simulation results of Estrada-Allis et al. (2019) near the Canary Islands, although a detailed temporal analysis remains beyond the scope of our current study. It should be noted that, in our compositing procedure, the snapshots were not aligned according to the sign of the dipole. Consequently, for the snapshots where the dipole phase undergoes evolution or rotation throughout the lifetime, the vertical velocity signals will cancel out across different azimuthal directions during the averaging process, ensuring that the resulting composite isolates the most fundamental and persistent dipole structure. Besides, we note that Mason et al. (2017) performed a composite analysis by rotating eddies to align with their respective propagation directions. The results show that eddies of the same polarity produced consistent dipole patterns in different subregions of the Brazil Malvinas Confluence, and their spatial structure is consistent with the patterns identified in the subtropical region of our study. Their method effectively isolated the dipole signals related to propagation direction. In our composite analysis, our objective is to characterize the most fundamental structures in different regions, so the dipoles within a given region are averaged regardless of their individual propagation directions. This methodological choice is supported by our preliminary statistical analysis, which indicate that dipole structures dominant in both the Subtropical and Extension regions. We anticipate that the fundamental structure of resulting composite vertical velocity will consistently exhibit a dipole structure, a hypothesis confirmed by our final results (where we observed that even snapshots initially categorized as monopole or quadrupole modes often exhibit a dipole pattern in the composite, although these results are not shown in the current manuscript). In fact, within our analysis framework, the effect of propagation direction is included in the role of the regional background flow. Fundamentally, these two perspectives converge on the same physical mechanism: the dynamical impact induced by the background flow on the eddy structure. Our approach deepens the understanding of this interaction by situating the eddies within the specific regional dynamical contexts.

Furthermore, the upper ocean is characterized by submesoscale dynamics, which can also strongly enhance the vertical motion. These motions can manifest as a pattern of alternating filaments with velocities an order of magnitude greater than those in the thermocline (as shown in Figure 4). However, the intense instability and rapid spatiotemporal variability of submesoscale motions may have complex implications for primary production in the euphotic layer. Therefore, it is necessary to consider the consistency of the timescales between vertical transport and primary production (Chen et al., 2020). Moreover, according to Chang et al. (2017), the cells within multipole structures can have opposing effects on nutrient transport. It should be noted that investigating such transport dynamics requires a focus on the material trapping and leakage properties of coherent eddies, for which a Lagrangian tracking framework, rather than the Eulerian approach adopted in this study, might be a more suitable choice. According to Liu & Abernathy (2023), the Lagrangian eddies generally have smaller radius and mainly represent the coherent core of Eulerian eddies, therefore we expect that it may impact the intensities of the results. Since the eddy classification method in this study is based on the instantaneous eddy snapshots, the composited results based on the Lagrangian tracking method may exhibit some differences. However, given that the main difference between the two identification methods lies in the eddy radius, we expect that the fundamental composited dipole structure would remain. Given this complexity, the actual material trapping/leakage and net transport of upwelling and downwelling cells should be considered when evaluating their importance in biogeochemical processes. Therefore, the multipole structures of vertical velocity and their impacts on ecological processes require further investigation.

The prevailing understanding suggests that vertical velocities within oceanic mesoscale eddies are dominated by monopole structures, with upwelling (downwelling) occurring in cyclonic (anticyclonic) eddies in the Northern Hemisphere. Our results show that dipole patterns dominate the vertical velocity structures within mesoscale eddies. The combined influence of the β -effect and the deformation effect related to background currents modulates opposing dipole patterns across different regions. Similar influence of ageostrophic processes is also noticed in an anticyclonic eddy near the Canary Islands by Estrada-Allis et al. (2019), yet they did not emphasize the combined effect of the above two terms. These findings improve our understanding of the three-dimensional motions within mesoscale eddies. Owing to rapid spatial and temporal variability, direct measurements of vertical motions remain challenging; thus, high-resolution numerical simulation is still the primary approach for us to understand vertical motions within mesoscale eddies. In this study, we employ the idealized model to focus on the dynamical mechanisms underlying the dipole structures and their regional differences. By simplifying the model configuration that avoid additional processes such as topography, the resulting dipole mechanisms can be identified and interpreted more clearly. Considering the important role of mesoscale eddies in redistribution of

nutrients, oxygen, and organisms in the upper ocean, the dipole structures may also play a crucial role in driving the biogeochemical extremes within the euphotic layer. These aspects will be further investigated in future works. Additionally, we reveal a stronger dipole vertical velocity of eddies in the extension region, implying broader applicability of the proposed mechanism in regions with strong eddy-background current interactions, such as the Southern Ocean and the Gulf Stream. These findings may have important implications for understanding vertical exchange in mesoscale eddies and their ecological consequences across different regions.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Availability Statement

The study is based on outputs from numerical simulation, the scripts for eddy identification, statistical analysis, composite analysis, and diagnostic analysis are available online and preserved in the zenodo repository (https://zenodo.org/account/settings/github/repository/FuncUny/Eddy_wvelDipole, <https://doi.org/10.5281/zenodo.20032971>).

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